Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Development of a Next-Generation AI-Agent Driven Enterprise Imaging Platform with Extreme SLM Compression and Multi-Modal Radiogenomics Integration			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Information & Systems	Other Medical Information & Systems	Software	Software Solutions

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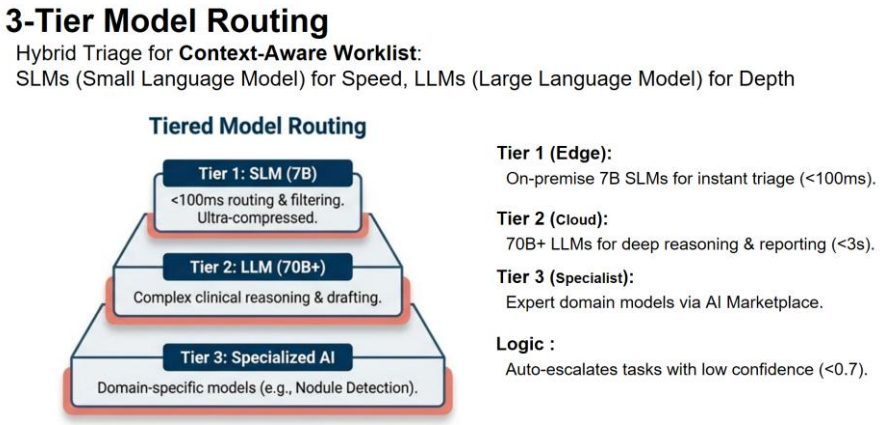
Concept of Proposed Technology

The targeted platform is an autonomous, AI-driven Enterprise Imaging solution designed to fundamentally shift radiology workflows from passive "viewing" to active "agentic collaboration." To ensure real-time responsiveness and data security, targeted platform operates on a hierarchical **3-Tier Intelligence Architecture (Tier 1 Local Edge SLM ↔ Tier 2 Cloud based LLM ↔ Tier 3 Domain Specific models)**.



IAP 3-Tier Model routing concept diagram

Modern radiology workflows generate thousands of examinations daily, requiring intelligent prioritization. Current 7B SLM inference exhibits latency of approximately 976ms at batch size 1 [1], far exceeding the <100ms requirement for seamless clinical integration. While 4-bit quantization reduces memory by 60-70% and improves inference speed by 2-3× on edge devices [2], existing joint pruning-quantization methods such as SparseGPT [3] suffer from catastrophic quality degradation at extreme compression.

The root cause is that Standard normalization techniques erroneously include structural zeros (from pruning) in variance computations, systematically deflating weight distribution statistics. At 50% sparsity, variance is underestimated by a factor of 0.5, causing quantization scales to be $\sqrt{2} \approx 1.41\times$ too small—catastrophic at 3-bit precision (only 8 quantization levels).

To address these fundamental statistical incompatibilities and performance bottlenecks, this proposal specifically focuses on the Tier 1 (Local Edge) layer, which serves as the "reflexive nervous system" of the platform6. By conducting joint research on Extreme SLM Compression, we aim to integrate Tier 1 SLMs directly into the on-premise infrastructure. This enables the Tier 1 engine to proactively manage clinical workloads—such as instant triage and worklist prioritization—with zero latency, independent of heavy cloud connectivity. Critically, Tier 1 also serves as the intelligent routing layer that determines when cases require escalation to Tier 2's cloud-based LLM for complex reasoning tasks—such as multi-study correlation or nuanced differential diagnosis—or to Tier 3's domain-specific models for specialized analyses like radiogenomic phenotype matching.

Core Functions & Detailed Technologies:

1. Agentic Workflow Engine:

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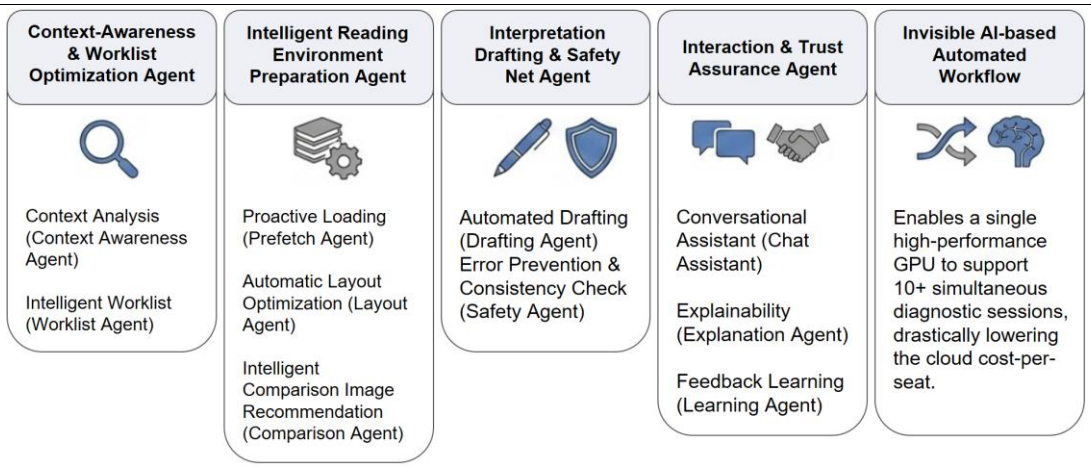

Targeted Core Service Agents - Implemented through the organic collaboration of the following detailed agents.

- **Agentic Workflow Engine (Multi-Agent Orchestration):** Based on the Tier 1 Intelligence Architecture (Local Infrastructure & Edge Core), the system operates autonomously through a continuous "Perception → Reasoning → Action" loop, utilizing specialized agents for each stage of the diagnostic workflow.

- **Worklist Agent (Real-time Triage):** Situated on the on-premise edge, this agent analyzes incoming exams with <0.1s latency per case. It acts as a "Medical Triage Officer," instantly filtering and prioritizing critical cases (e.g., ICH, PE) in the worklist before they reach the cloud core.

- **Drafting Agent (Pre-writing):** Instead of waiting for user dictation, this agent proactively generates Standard Reports and diagnostic drafts. It synthesizes imaging findings with patient history using RAG (Retrieval-Augmented Generation), providing a "Pre-written" report the moment the radiologist opens the study.

- **Safety Agent (Hallucination & Consistency Check):** Operates in the background to monitor the report in real-time. It cross-references the "Findings" and "Impression" sections to detect logical inconsistencies (e.g., ensuring a described lung nodule is noted in the conclusion) and prevents AI hallucinations, serving as a final quality control layer.

2. Tier 1 SLM Intelligence (Extreme Compression at the Edge):

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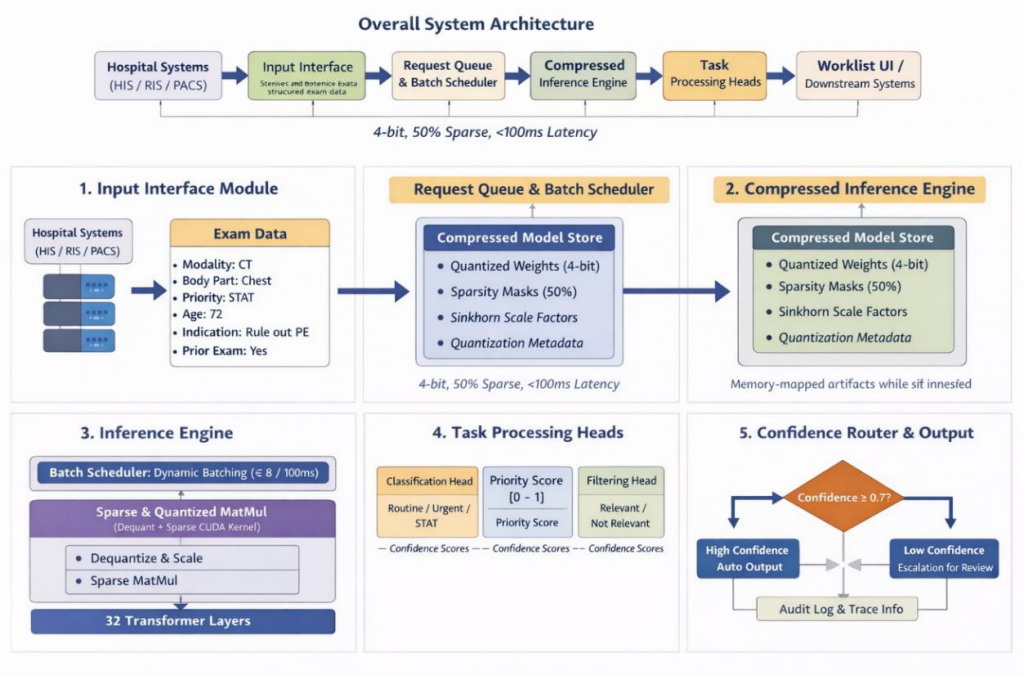

System architecture for ultra-low-latency clinical workflow automation using an extremely compressed small language model.

- **Technology:** Deployment of **7B-parameter Small Language Models (SLMs)** compressed to **4-bit precision with 50% sparsity** on commodity edge/consumer grade GPUs (e.g., RTX 4090).

- **Innovation:** Utilizes **"Sparse-Aware Sinkhorn Normalization"** to solve the 'variance collapse' problem in joint pruning-quantization, achieving **<100ms inference latency** (vs. 976ms baseline) with **<2% perplexity degradation**.

- **Application:** Acts as a real-time "Medical Triage Officer," instantly filtering thousands of incoming exam orders for urgency (e.g., Stroke, PE) and reorganizing worklists based on clinical context before the radiologist even opens the system. Beyond autonomous triage, the Tier 1 SLM generates structured metadata and confidence scores that are passed upstream to Tier 2 and Tier 3 when the clinical complexity exceeds local inference thresholds—ensuring seamless handoff while preserving the sub-100ms response time for routine cases.

3. Hyper-Stream Cloud Architecture (SSR):

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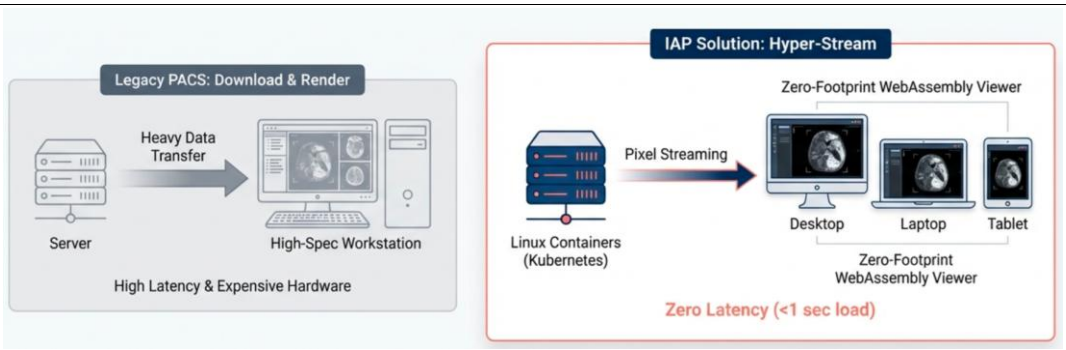
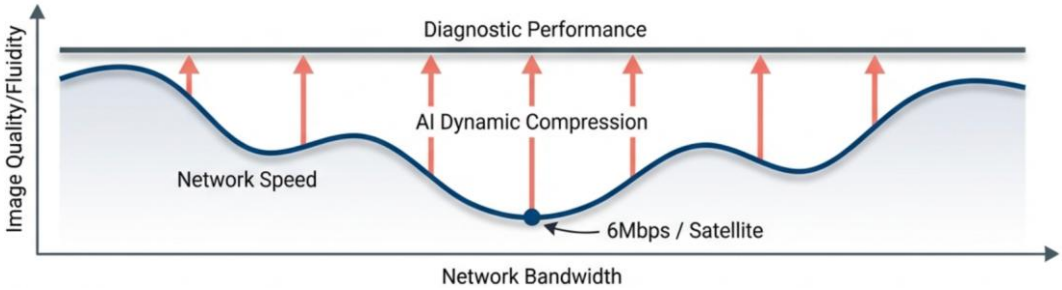
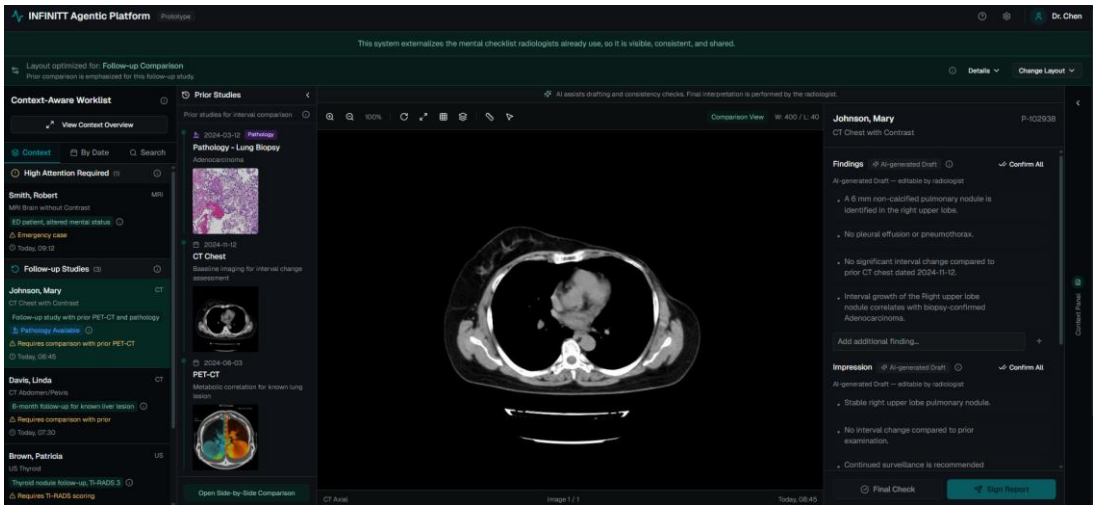


- **Zero-Footprint:** A 100% Server-Side Rendering (SSR) viewer that streams pixels instead of data, enabling **<1 second image loading** for massive 3D/4D datasets (e.g., 10,000+ slice CTs).



- **AI-Adaptive Streaming:** Deep learning models predict network conditions to dynamically adjust compression, ensuring smooth diagnosis even in low-bandwidth (6Mbps) or satellite environments.

4. Multi-Modal Radiogenomics (ULiteX):



- **Integration:** A unified dashboard correlating **Radiology** (Imaging phenotypes), **Pathology** (WSI), and **Genomics** (NGS variants) data.

- **Insight:** Enables "Phenotype-Genotype Mapping," where clicking a gene mutation (e.g., EGFR) highlights the corresponding tumor region in the MRI/CT.

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Rationale for Joint R&D Project	**1. Technological Leap (Overcoming SLM Limits):** - Current SLM inference is too slow (approx. 1 sec) for real-time clinical workflows. To achieve sub-100ms latency without losing diagnostic accuracy, breakthrough compression algorithms are required. - **University of Toronto** possesses world-class expertise in **sparse matrix optimization** and **quantization theory**, essential for co-developing the "Sparse-Aware Sinkhorn Normalization" algorithm which is the key enabler for our Tier 1 SLM. **2. Clinical & Regulatory Validation:** - **Medical Center (MC):** As a leading global hospital in precision medicine, MC provides the critical **multi-modal datasets (Radiology + Pathology + Genomics)** needed to train and validate the Radiogenomics engine. Their participation ensures the platform meets the rigorous clinical standards required for high-stakes cancer diagnostics. **3. Global Market Expansion:** - This collaboration secures both the "Deep Tech" (Univ. of Toronto) and "Clinical Trust" (MC) necessary to compete with global incumbents like Visage and Sectra in the North American and Asian markets.
Key Words	1. Agentic AI & Autonomous Workflow 2. Extreme SLM Compression (Sparse-Aware Sinkhorn) 3. Hyper-Stream Cloud (Server-Side Rendering) 4. Radiogenomics (Phenotype-Genotype Integration) 5. Next-Gen Enterprise Imaging

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (8)
R&D Objectives	The primary objective is to develop and commercialize an agentic platform, a "Self-Driving" workflow engine that reduces radiologist burnout and enables precision medicine. **Detailed Technical Objectives:** **1. Extreme SLM Optimization:**			

- Develop a **Joint Pruning-Quantization** pipeline achieving **8x compression** (14GB → 1.75GB).
- Implement **Sparse-Aware Sinkhorn Normalization** to maintain statistical fidelity in 50% sparse matrices.

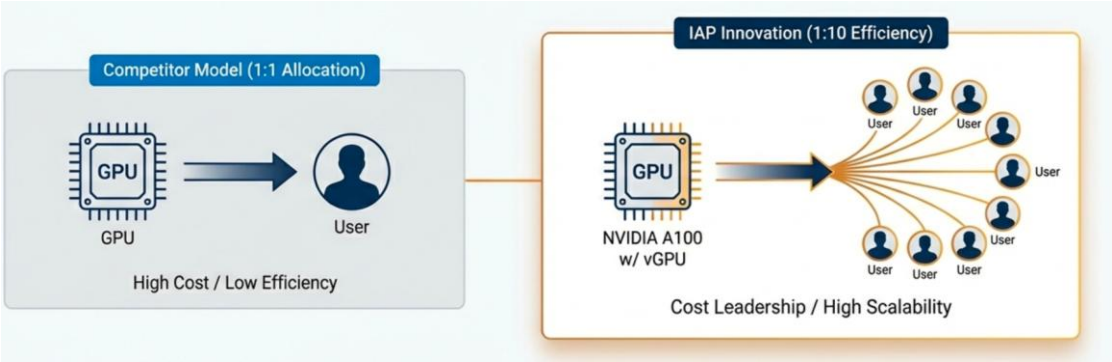
$$\text{Sparse-Aware Variance:} \quad \sigma^{\tilde{2}} = [\Sigma(M_{ij} \cdot W_{ij}^2) \;/\; \Sigma(M_{ij})] - [\Sigma(M_{ij} \cdot W_{ij}) \;/\; \Sigma(M_{ij})]^2$$

$$\text{Importance Score:} \quad I_{ij} = |W_{ij}| \cdot \|X_j\|_2 \;/\; (\mu_{1j} \cdot \mu_{2i} + \varepsilon)$$

Compression Algorithm

- Achieve **<100ms inference latency** (Batch=1) on consumer-grade GPUs. This extreme compression target is architecturally essential: by reducing Tier 1's memory footprint to <2GB, we free GPU headroom to run lightweight adapter layers that format output for Tier 2's cloud LLM context window and Tier 3's domain-specific model input schemas, enabling unified inter-tier communication protocols.

2. Intelligent Cloud Infrastructure:



- Build a Linux-containerized SSR engine with **GPU Virtualization (vGPU)** to support 10+ sessions per A100 GPU.
- Develop **AI-based Predictive Prefetching** to achieve "Zero-Click" access times.

3. Integrated Diagnostics:

- Create the **ULiteX** viewer capable of rendering WSI (Pathology) and NGS (Genomics) tracks alongside DICOM images.
- Develop specific agents for **Smart Lesion Tracking** and **Automated Report Drafting**.

R&D Target Specifications

1. SLM Performance (Tier 1 Intelligence):

- Model Size: **< 2GB** (Targeting 7B parameter models).
- Latency: **< 100 ms** per case classification.
- Accuracy Preservation: **< 2% increase in Perplexity (PPL)** vs. FP16 baseline.
- Throughput: **> 1,000 exams/hour** on a single edge node.

2. System & Cloud Performance:

- Image Load Time: **< 1 second** (Average across all modalities).

	- Bandwidth Efficiency: Usable at **6 Mbps**. - System Downtime: **50% reduction** via Self-Healing agents. **3. Clinical Efficiency:** - Report Drafting Time: **50% reduction**. - User Clicks: **50% reduction** via context-aware automation.
R&D Plan	**Year 1 (2026): Core Technology & SLM Algorithm** **[Korean SME]** Launch AWS Cloud PACS with basic Hyper-Stream protocol; Collect baseline workflow data. **[Univ. of Toronto]** Develop theoretical framework for "Sparse-Aware Sinkhorn Normalization"; Implement baseline joint pruning-quantization on LLaMA-7B. Concurrently, define the API contract specifying how compressed Tier 1 inference outputs (confidence scores, entity extractions, escalation flags) will interface with Tier 2's cloud orchestration layer and Tier 3's domain model endpoints. **[MC]** Define clinical requirements for Radiogenomics; Prepare de-identified multi-modal datasets. **Year 2 (2027): System Integration & Validation** **[Korean SME]** Integrate compressed SLM engine into IAP Edge Core; Develop Beta version of ULiteX viewer. **[Univ. of Toronto]** Optimize CUDA kernels for sparse matrix multiplication; Benchmark on diverse hardware (GPU based). Validate that the optimized sparse kernels maintain output compatibility with Tier 2/Tier 3 integration protocols under varying network latency conditions, ensuring graceful degradation when cloud connectivity is intermittent. **[MC]** Pilot deployment of IAP Beta; Validate Radiogenomics dashboard with real cancer cases. **Year 3 (2028): Commercialization & Expansion** **[Joint]** Finalize "v1.0" for global launch; Preparing FDA 510(k) Clearance. **[Korean SME]** Launch Global SaaS service; Open "AI Marketplace" for 3rd party agents. **[MC]** Publish clinical study results on AI-driven workflow efficiency and Radiogenomics utility.

4. R&R (Roles and Responsibilities)

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Expected R&R of Research Institute	**Korean SME Healthcare R&D Center** 1. **Project Management & System Integration:** Overall architecture design, cloud platform development, and commercialization strategy. 2. **AI Agent Development:** Development of Drafting, Safety, and Layout agents; integration of Tier 1 SLM into the entire workflow. 3. **Productization:** UI/UX design (ULiteX), regulatory compliance (FDA/CE), and global sales execution.
Expected R&R of Korean SME Partner	**1. University of Toronto (Academia):** - **Algorithm Development:** Research and develop the "Sparse-Aware Sinkhorn Normalization" technique to correct variance shifts in quantized sparse matrices. - **Optimization:** Create custom CUDA kernels for efficient execution of 4-bit sparse models on commodity hardware. - **Benchmarking:** Validate compression performance across various SLM families (LLaMA, Qwen, Gemma). Additionally, collaborate with Korean SME to design the Tier 1→Tier 2 escalation heuristics, ensuring that compressed model outputs contain sufficient semantic fidelity to inform cloud-based LLM reasoning without requiring raw data retransmission. **2 Medical Center (Medical Partner)** - **Radiogenomics Data:** Provide curated, de-identified datasets linking Imaging (MRI/CT), Pathology, and Genomics (NGS). - **Clinical Pilot:** Serve as the primary testbed for IAP; Evaluate the clinical utility of AI agents in a high-volume tertiary hospital setting. - **Medical Advisory:** Define protocols for "Integrated Diagnostics" and provide feedback on AI safety/consistency. **3. Korean Medical AI SMEs (Potential Partners):** - Joint development of specialized Medical AI modules (e.g., specific disease detection models) to be onboarded onto the entire system. - Verification of interoperability with diverse systems.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**1. Technological Impact:** - Establishment of a "First-Mover" position in the agentic AI radiology market, surpassing current "AI-as-a-Tool" paradigms. - securing core IP for extreme LLM/SLM compression, enabling high-performance AI on low-cost edge hardware. **2. Commercial Potential:** - Shift from one-time license sales to a **SaaS/Subscription model**, targeting 60%+ overseas revenue share by 2030.

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	- Expansion into the high-value Precision Medicine market via Radiogenomics. **3. Economic & Industrial Impact:** - Strengthening the competitiveness of the Korean medical SW industry ("K-Medical"). - Cost reduction for hospitals by utilizing commodity GPUs (vs. expensive Enterprise GPUs) and reducing IT maintenance via Self-Healing agents.

6. References

[1] Clarifai, "LLM Inference Optimization Techniques," 2025. Benchmark: 7B model latency 976ms at batch=1.

[2] EdgeProfiler, "A Fast Profiling Framework for Lightweight LLMs on Edge," arXiv:2506.09061, 2025.

[3] E. Frantar and D. Alistarh, "SparseGPT: Massive Language Models Can Be Accurately Pruned in One-Shot," ICML, 2023.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization